# Mayur Garg

**Male, 29 years old**

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Research-focused individual currently working as a Data Scientist at United Airlines with 7+ years of experience in Python, SQL, Machine learning and AI.

## PROFESSIONAL EXPERIENCE

### **United Airlines,** Gurugram

Data Scientist : JULY 2024 - PRESENT

- Worked in the Analytics and Innovation team to solve business use cases using machine learning.

**Key Projects**

- Worked on forecasting base fare prices for other airlines in advance to be used as an input for the revenue management team.
- Classified invoices accounting for spend of over $50B every year into a multi-level hierarchy allowing the team to gain visibility into spending patterns and manage supplier relations via a PowerBI dashboard.
- Worked in a team to deliver a multi-agent LLM solution for quick retrieval of knowledge documents and faster resolution for incident, change and problem workflows.

### **Samsung Research Institute,** Bengaluru

Senior Engineer (Data Scientist) : OCTOBER 2022 - JUNE 2024

- Worked in a research-focused team with initiatives in generative AI, time series and recommender systems to complement various Samsung applications.

**Key Projects**

- Implemented a solution that forecasts the demand of products at a warehouse level for retail beating the existing methodology in multiple benchmarks.
- Created a dashboard using BigQuery ML that forecasted the optimal pricing of multiple Samsung retail products to maximize both profit and revenue.
- Co-created a chat assistant using generative LLMs for information retrieval and conversational recommendation for Samsung smartphones.
- Implemented a web app to generate personalized product information and marketing promos by analyzing user queries to assist tech support in real time.
- Authored and published two research papers.

**Rewards and Recognition**

- Awarded the ‘Spot Award’ in Aug ‘23 for fine-tuning LLMs on custom chats.
- Cleared the Advanced and Professional levels of Samsung’s Software Competency Assessment.

### **Deloitte USI,** Gurugram

Consultant (Developer) : JULY 2021 - SEPTEMBER 2022
Business Technology Analyst (Developer) : JULY 2019 - JUNE 2021

- Worked in a team of 100+ people to develop web applications which were used to

## PROFILE LINKS

**LinkedIn**

**GitHub (mayur7garg)**

**Kaggle (mayur7garg)**

**Google Scholar**

## TECHNICAL SKILLS

**DBMS**
MS SQL Server
MySQL
Redis
SQLAlchemy
BigQuery

**Python modules**
numpy
pandas
polars
scikit-learn
Tensorflow + Keras
PyTorch
transformers
NLTK
Gensim
xgboost

**Frameworks**
LangChain
FastAPI

**Cloud and Collaboration**
Git
Jira
Azure Devops

**Other Tools**
Docker
Postman
Rabbit MQ

provide surveying and analytic services for tax credit applications.

**Key Projects**

- Implemented an NLP web app that allows the tax admins to view, highlight and annotate useful text snippets in documents with key phrases automatically tagged using key phrase extraction, semantic matching and custom heuristics.
- Led a project that allows admins to develop customizable surveys for quantifying tax credits for R&D activities based on client needs that uses a data driven approach instead of the pre-existing code driven approach.
- Implemented an NLP text classification POC that assigns emails from multiple inboxes to appropriate teams based on their content and resource availability.

**Rewards and Recognition**

- Stood 4th among 30+ teams in an intra Deloitte ML hackathon.
- Awarded the 'Applause Award' in Nov '21 as part of Deloitte's R&R initiative.

## PUBLICATIONS

**A Distribution Agnostic Rank-Based Measure for Proximity Search** — *IEEE Access Volume 13, Dec 2025*

**Publication** - https://ieeexplore.ieee.org/document/10815932

**News Popularity Beyond the Click-Through-Rate for Personalized Recommendations** — *SIGIR'23, July 2023*

**Publication** - https://dl.acm.org/doi/10.1145/3539618.3591741

**Analysis of Genetic Algorithm in Infinite Monkey Problem** — *IJSRD Volume 9, Issue 7, Oct 2021*

**Publication** - http://ijsrd.com/Article.php?manuscript=IJSRDV9I70023

## EDUCATION

**Indian Institute of Technology (IIT-D),** Delhi — *MS(R) (Part time)*
JULY 2025 - Ongoing
**Dept.** - Computer Science and Engineering (CSE)

**Maharaja Agrasen Institute of Technology (IPU),** Rohini — *B.Tech.*
AUGUST 2015 - JUNE 2019
**Field** - Computer Science and Engineering (CSE) | **CGPA** - 8.49/10

**New Green Field School,** Alaknanda— *CBSE XII*
APRIL 2014 - MARCH 2015
**Field** - PCMB | **Aggregate** - 90.6%

**New Green Field School,** Alaknanda— *CBSE X*
APRIL 2012 - MARCH 2013
**CGPA** - 10.0/10

## CERTIFICATIONS

**DP-900: Microsoft Certified - Azure Data Fundamentals**
**Issued on** - Feb 13th, 2022

## MACHINE LEARNING

**Supervised**
Linear Regression
Logistic Regression
Decision Trees
Random Forests
SVM
Naive Bayes
KNN

**Unsupervised**
KMeans
DBSCAN
PCA

**Deep Learning**
CNNs
LSTMs
Autoencoders
Diffusion
Transformers

**NLP**
Topic Modeling
Keyphrase Extraction
Text Generation
LLMs
Agentic AI

## OPEN SOURCE PROJECTS

Feistel Cipher PyPI package

Android app with multiple implementations of Genetic algorithm

Stealth Android Game